MUTHU NANTHINI P

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PROFESSIONAL SUMMARY

Final-year B.E. Computer Science and Engineering student (CGPA: 8.23, graduating 2027) with hands-on experience in software development, web development, API integration, and database management. Built SmartHMS, a role-based hospital management system using Spring Boot, Python FastAPI, React.js, and MySQL; completed three internships. Skilled in debugging, problem-solving, and Git.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL
Frontend Development: React.js, Angular, HTML5, CSS3, Tailwind CSS, Chart.js, Recharts, Axios
Backend Development: REST APIs, Python FastAPI, Spring Boot, Spring Data JPA, Spring Security, JWT Authentication, Maven
Database: MySQL
Tools & Technologies: Git, GitHub, Postman, VS Code, IBM Cognos Analytics
Technical Concepts: API Integration, CRUD Operations, Dashboards, Agentic AI Workflows, Responsive Design, Debugging & Testing
Spoken Languages: English, Tamil

PROJECTS

SmartHMS (MediCore AI) - AI-Powered Hospital Management System
Java 21, *Spring Boot 3*, *Spring Data JPA*, *Maven*, *Spring Security*, *JWT*, *React.js*, *Tailwind CSS*, *Chart.js*, *MySQL*, *Python FastAPI*

- Built REST APIs with Spring Boot and Python FastAPI covering 7 hospital modules (patients, appointments, prescriptions, lab tests, pharmacy, billing, beds); tested endpoints using Postman.
- Secured the application with JWT and Spring Security role-based authorization for 6 user roles (Admin, Doctor, Receptionist, Lab Staff, Pharmacist, Patient).
- Developed role-based React.js dashboards with Tailwind CSS and Chart.js analytics, integrated with the APIs using Axios.
- Designed a normalized (3NF) MySQL database with 13+ entities, reducing data duplication across the hospital workflow.

Butterfly Jewel - E-Commerce Jewellery Shopping Platform
HTML, *CSS*, *JavaScript*, *Java*, *Spring Boot*, *MySQL*

- Built a responsive e-commerce storefront with JavaScript-driven live cart updates (no page reloads) and client-side form validation to keep user input clean.
- Integrated Spring Boot REST APIs via fetch/AJAX for authentication, products, cart, and order processing, storing data in MySQL.

AI Student Support Assistant - IBM Agentic AI Internship Project
React, *TypeScript*, *Tailwind CSS*, *Recharts*

- Built a student support web app with an AI chat assistant that answers academic questions, cites sources, and explains how each answer was generated.
- Implemented an agentic workflow that interprets queries, selects tools, and retrieves information from college documents; delivered a frontend prototype ready for LLM/RAG backend integration.

INTERNSHIP EXPERIENCE

TVK Technologies Jun 2026 - Jul 2026
Java Full Stack Development Intern | Chennai, Tamil Nadu (Remote)

- Built Hospital Management System modules using Java, Spring Boot, MySQL, HTML, CSS, and JavaScript, connecting frontend screens to REST APIs through project assignments.

NEXDHA AI Fintech Pvt. Ltd. Dec 2025 - Jan 2026
Web Development Intern | Chennai, Tamil Nadu (On-site)

- Built responsive web pages and user interfaces as part of the development team, translating business requirements into features, with debugging and testing.
- Gained hands-on backend development skills and applied them to build the Butterfly Jewel e-commerce platform.

Adroit Technologies Innovative Solutions Pvt. Ltd. Mar 2026 - Apr 2026
Virtual Intern - IBM Cognos Analytics

- Built an academic analytics dashboard visualizing pass/fail results, enrollment trends, and grade distribution, with KPI tiles for student count, attendance, and placement status.

EDUCATION

B.E. in Computer Science and Engineering | V V College of Engineering, Tisaiyanvilai | CGPA: 8.23 2023 - 2027
HSC (12th Standard) | Valliammaiyar Girls Hr. Sec. School, Kulasekarapatnam 70.17%
SSLC (10th Standard) | Valliammaiyar Girls Hr. Sec. School, Kulasekarapatnam 100%

CERTIFICATIONS

- **NPTEL Online Certification (Elite): Cloud Computing**, IIT Kharagpur (12-week course, Jul-Oct 2026), consolidated score 66%.
- Internship Certificates: Java Full Stack (TVK Technologies); Web Development (NEXDHA AI FINTECH Pvt. Ltd.); IBM Cognos Analytics Virtual Internship (Adroit Technologies).
- Spoken English Course (Project Punch), Beehive Communication Club, V.V. College of Engineering, Mar 2026.
- Symposium Participation: NEXUS-25, Grace College of Engineering (Oct 2025); THINKSTRA '26, Dr. Sivanthi Aditanar College of Engineering (Mar 2026).